ETF TSMOM 12-Month

Strategy report · Research desk · 09 November 2026

Status: passed, confirms test zone · *full-sample t 2.16*

Abstract

ETF TSMOM 12-Month goes long or short each of 22 ETFs (equities, bonds, commodities, currencies) according to the sign of its own past 12-month excess return, risk-balanced to 8 % volatility and rebalanced monthly. Over January 2006 to September 2026 it earns a net Sharpe ratio of 0.45 (90 % bootstrap range 0.11 to 0.81), with no exposure to SPY. On the test, which runs from 2021 and was opened once, the net Sharpe is 0.57 (range −0.06 to 1.20). **Three findings weigh against it: the short leg loses money in every period, the whole profit sits in a few days (without the best 1 % of days the full-sample Sharpe is −0.26), and the validation period, 2018 to 2020, lost money.** The test confirms the earlier result and the strategy is kept as passed, with a thin edge.

1 Hypothesis and method

Investors under-react to news, then chase the move once it is visible, so a price that has risen for twelve months tends to keep rising. In commodities, hedgers also pay speculators for liquidity through the roll. The pattern comes from Moskowitz, Ooi and Pedersen (2012), measured in-sample on 58 futures from 1985 to 2009, before costs, and from Lempérière and others (2014), who report a gross Sharpe of 0.72 for trend since 1800. Applying the same rule to 22 ETFs instead of futures is the desk's own construction.

The universe is 22 ETFs in four classes: nine equity, five bond, five commodity and three currency. An ETF enters after 12 months of history. The list is 22 survivors picked in 2026, so every result is conditional on that list. Each month end the book takes the sign of the 12-month excess return over the T-bill rate, sizes each ETF by the inverse of its volatility, and scales the book to 8 % ex-ante volatility with a gross cap of 2. Costs are 2 bp on ten liquid ETFs and 5 bp on the other twelve, plus 25 bp a year of borrow on the short gross. The train and the validation (the three years from 2018) were used to design and check the rule; the test starts in 2021 and was opened once, on 9 November 2026.

2 Performance

The full sample includes the train, on which the strategy was designed; the test is the honest estimate. Over the test the strategy returned 4.4 % a year for 7.8 % volatility, a net Sharpe of 0.57, and the bootstrap range of −0.06 to 1.20 cannot tell it from zero. The full sample gives 3.6 % a year and 0.45. By period the Sharpe is 0.54 on the train, −0.14 on the validation and 0.57 on the test, so the edge disappeared for three years and came back. The worst years are 2016 (−12 %) and 2023 (−6 %); the best are 2013 (15 %) and 2022 (12 %). The maximum drawdown, −17.5 %, runs from October 2022 to April 2025 and was recovered in September 2026; the longest time under water is 1486 days.

	Sample	5th percentile	Median	95th percentile
Annualised return	3.6 %	0.9 %	3.6 %	6.4 %
Annualised volatility	8.0 %	7.6 %	8.0 %	8.4 %
Sharpe ratio	0.45	0.11	0.45	0.81
Maximum drawdown	−17.5 %	−38.1 %	−21.8 %	−13.9 %
Worst month	−5.7 %	−8.6 %	−6.4 %	−5.0 %
Positive months	53 %	52 %	57 %	62 %

Table 1: Net performance, full sample, January 2006 to September 2026, after costs and financing. Percentiles of a stationary block bootstrap of daily returns (blocks of 20 days, 2 000 paths). Returns in fraction of capital, not compounded.

	Sample	5th percentile	Median	95th percentile
Annualised return	4.4 %	−0.5 %	4.4 %	9.3 %
Annualised volatility	7.8 %	7.1 %	7.8 %	8.4 %
Sharpe ratio	0.57	−0.06	0.57	1.20
Maximum drawdown	−17.5 %	−23.7 %	−13.0 %	−7.6 %
Worst month	−5.7 %	−7.2 %	−5.0 %	−3.4 %
Positive months	54 %	48 %	58 %	68 %

Table 2: Net performance over the test alone, January 2021 to September 2026. Same method as Table 1.

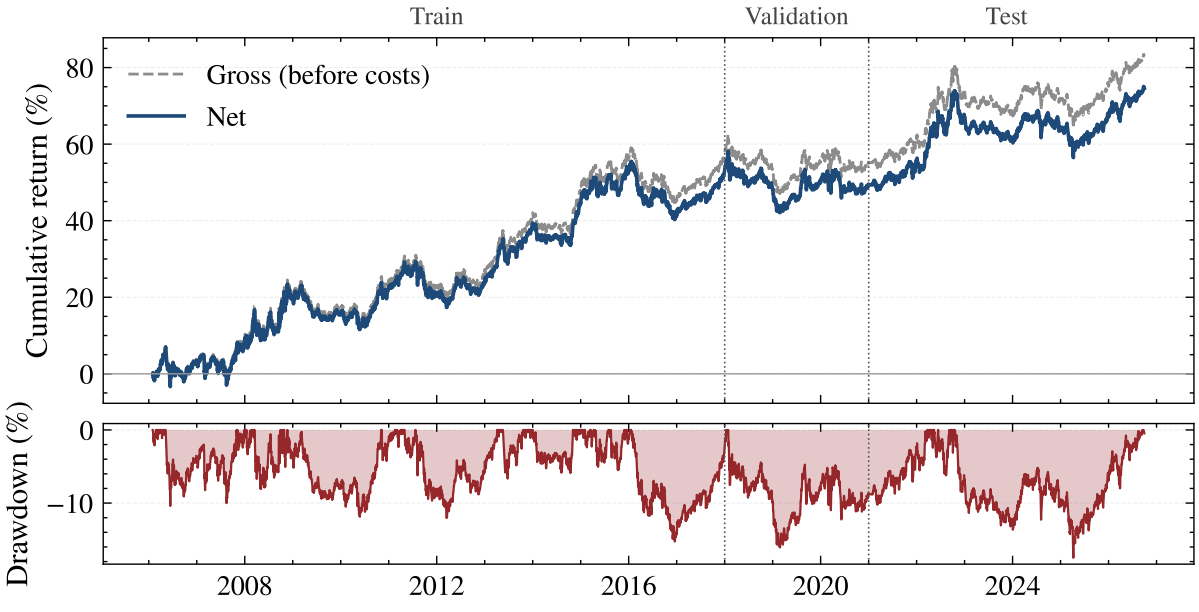


Figure 1: Cumulative net and gross return (top) and drawdown of the net return (bottom), in percent of capital, linear scale. Dotted lines mark the start of the validation and of the test.

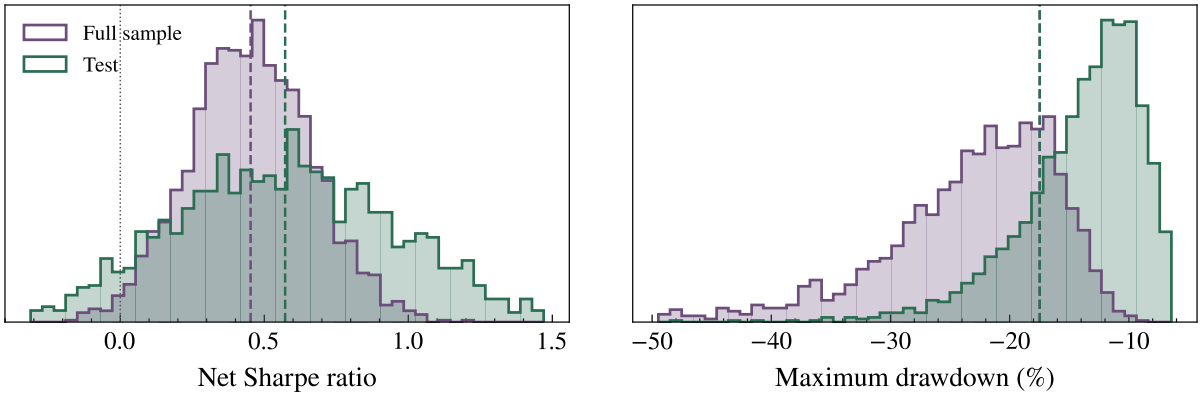


Figure 2: Bootstrap distributions of the net Sharpe ratio (left) and of the maximum drawdown (right). Dashed lines: the values measured on the sample.

3 Market exposure

The strategy is not a bet on the equity market. Its beta to SPY is 0.00 over the full sample and 0.09 over the test, and its correlation is 0.00 and 0.19. The alpha over SPY is 5.3 % a year on the full sample (t 2.09) and 4.7 % on the test (t 0.99), so it is shown on the long sample and not on the test alone.

	Full sample	Test
Beta to SPY (daily)	0.00 (t 0.06)	0.09 (t 1.83)
Correlation to SPY	0.00	0.19
Alpha over SPY, annualised	5.3 % (t 2.09)	4.7 % (t 0.99)

Table 3: Exposure to the market. Regression of daily net returns on SPY daily returns; t-statistics with Newey-West standard errors (10 lags).

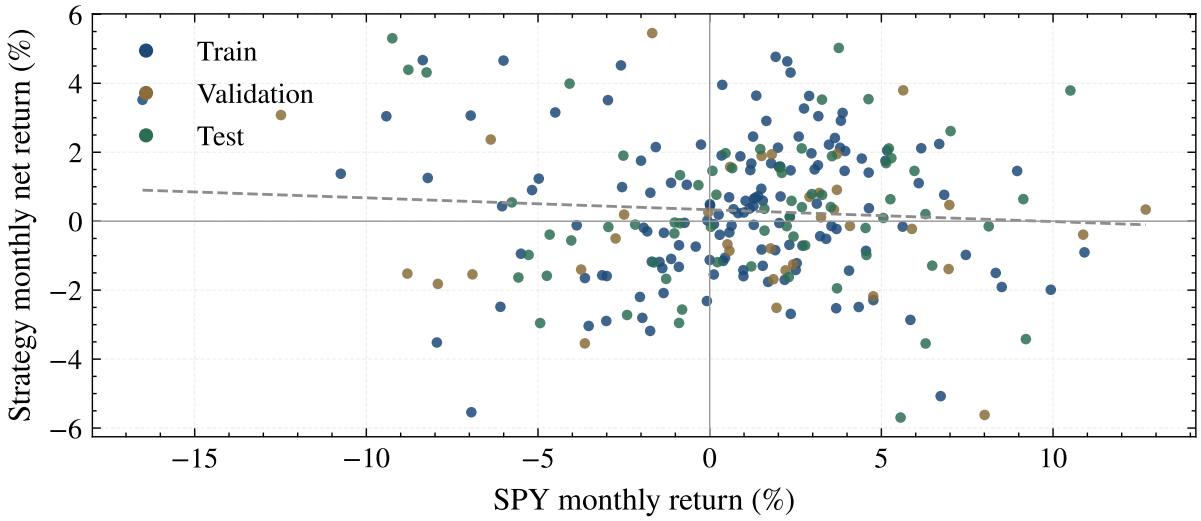


Figure 3: Monthly net return of the strategy against the monthly return of SPY. Dashed line: least-squares fit.

4 Where the profit comes from

The hypothesis says the profit comes from following the price, mostly on the long side of a market that trended up. It did: the long leg made 75 % of capital on the train, 11 % on the validation and 36 % on the test. The short leg lost in all three periods, −14 %, −12 % and −6 %. Financing at the T-bill rate cost 4 % on the train and 2 % on the test, and trading costs 4 %, 1 % and 3 %. The book is a long-biased trend book, and the short side adds nothing over twenty years. The payer named in the hypothesis paid, but only through the long leg.

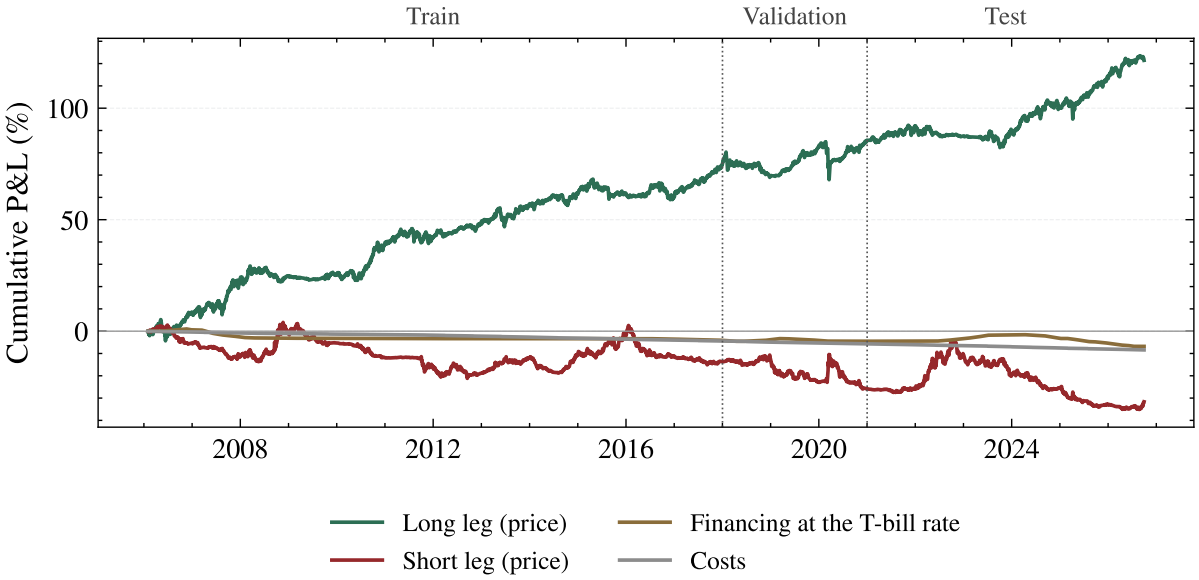


Figure 4: Cumulative P&L by source, in percent of capital: price of the long leg, price of the short leg, financing at the t-bill rate, and costs.

5 Concentration

The best 1 % of days make 152 % of the full-sample net P&L (76 days): without them the Sharpe falls from 0.45 to −0.26. On the test the best 21 days make 118 % of it and the Sharpe falls from 0.57 to −0.11. SPY, over the same sample, goes from a Sharpe of 0.77 to −0.01, so thick right tails belong to the market as much as to the strategy. The big days are regular: in every year the two best days of each month earn between 13 % and 33 %, while the other days lose between 3 % and 30 %, and without its two best days a month is positive only 27 % of the time. On names, the best five carry 52 % of the train P&L and 67 % of the test P&L, led by FXY on the test (16 %). Without the best five, the test keeps 8 % of capital.

	Full sample	Test
Net Sharpe ratio, all days	0.45	0.57
Share of net P&L from the best 1 % of days	152 % (76 days)	118 % (21 days)
Net Sharpe ratio without the best 1 % of days	−0.26	−0.11
Annualised return without the best 1 % of days	−1.9 %	−0.8 %
Net Sharpe ratio without the best 5 % of days	−2.17	−2.01
SPY Sharpe ratio, all days / without its best 1 %	0.77 / −0.01	

Table 4: Concentration of the gains in time. A day removed is set to zero, as if the strategy had been flat that day. The SPY line gives the same cut on the market over the same span.

	Train	Validation	Train and validation	Test
Names traded	22	22	22	22
Share of net P&L, best name	14 %	−24 %	14 %	16 %
Share of net P&L, best 5 names	52 %	−91 %	54 %	67 %
Share of net P&L, best 10 names	89 %	−138 %	84 %	107 %
Net P&L without the best 5 names, % of capital	25 %	−14 %	23 %	8 %

Table 5: Concentration of the gains across names: net P&L by name, best names first. A share above 100 % means the other names lost money together.

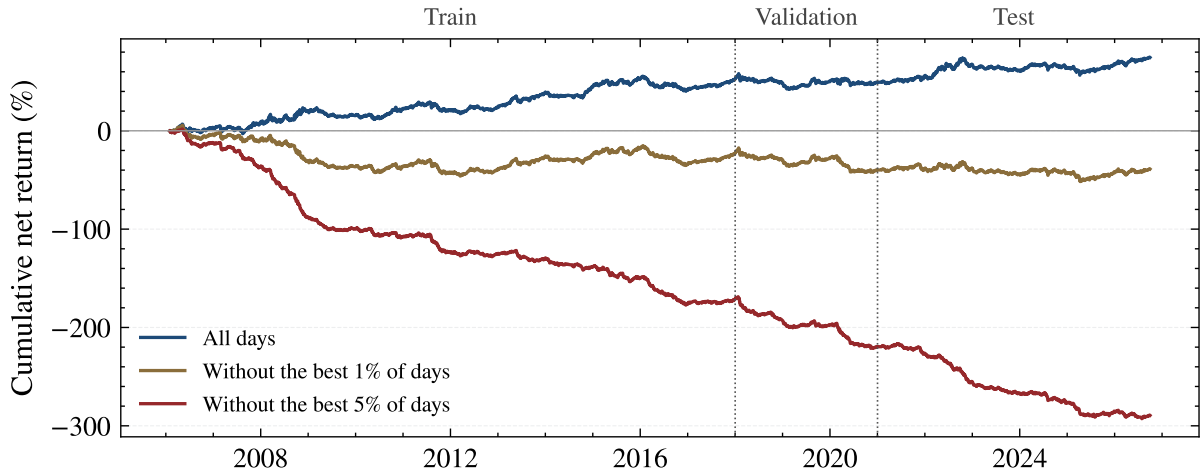


Figure 5: Cumulative net return with all days, and with the best 1 % and 5 % of days set to zero.

6 Implementation

The book holds about 13 names long and 9 short in the test, at a gross exposure of 1.63. Turnover is 7.5 a year and the cost drag is 0.5 % a year, in line with the train. The break-even cost is 73 bp on the train and 65 bp on the test, far above the 2 to 5 bp assumed. On the validation the break-even is −9 bp, because the strategy lost money before costs.

	Train	Validation	Test
Names held long / short, average	13 / 8	12 / 10	13 / 9
Gross exposure, average	1.54	1.86	1.63
Holding period of a position, days	4134	1096	2070
Turnover, times the capital a year	6.5	7.2	7.5
Cost drag, % of capital a year	−0.4 %	−0.5 %	−0.5 %
Break-even cost, bp one way	73.5	−8.7	65.3

Table 6: Implementation. The break-even cost is the one-way cost per unit traded at which the gross P&L of the period would be fully spent.

7 Robustness

The result survives costs: with costs doubled the train and validation Sharpe is 0.38 against 0.41 for the base, and with borrow at 100 bp a year it is 0.35. A one-day delay gives 0.35. The always-long control scores 0.36 and the twin control 0.13, so the base beats both but not by much, and the always-long book is a rival that is not far behind. Equal risk per class (variant A) scores 0.39 and does not replace the base. The nine ETFs launched after 2005 give 0.21, about half, with the sign kept; the 13 ETFs alive in 2005 give 0.42.

	Train and validation	Δ train and validation
Frozen version, as tested	0.41	
stress13	0.42	+0.01
stress9	0.21	−0.20
Costs × 2	0.38	−0.03
Borrow at 100 bp a year	0.35	−0.06
Trades 1 day later	0.35	−0.06

Table 7: Stress tests of the frozen version, run by the Red team before the test: net Sharpe ratio, and its change from the frozen version on the same period.

	Kind	Run on	Net Sharpe	Frozen version
control_long	control	Train	0.44	0.54
control_long	control	Train and validation	0.36	0.41
control_twin	control	Train	0.44	0.54
control_twin	control	Train and validation	0.13	0.41
A	variant	Train and validation	0.39	0.41

Table 8: Other versions run before the test, each on the period where the desk ran it: variants considered at the validation, controls and diagnostics that check the mechanism. They are not the strategy; the last column gives the frozen version on the same period for comparison.

8 Luck and multiple testing

Two versions of this strategy were run, and two for its family and asset, which behave as one independent try (rho 0.94). On 13.4 years the best of the tries with no skill would reach a Sharpe of −0.16, so the luck line is negative. The train net Sharpe of 0.54 clears it by 0.70, and the probability that its true value exceeds the line is 99 %. The line applies to the train, where the choices were made, and says only that the train result is not the best of many tries; it does not say the edge is large.

	Value
Tries counted for this strategy	2
Tries in the family, trend on multi-asset (effective number)	2 (1.0)
Luck line: best Sharpe of 1.0 tries with no skill, 13.4 years	-0.16
Train net Sharpe ratio / minus the luck line	0.54 / 0.70
Probability that the true train Sharpe exceeds the luck line	99 %
Whole desk: tries (effective number) / luck line	4 (1.4) / 0.05

Table 9: Multiple testing. Luck line: expected best annual Sharpe ratio of N_{eff} independent tries with no skill (Bailey and López de Prado, 2014); N_{eff} discounts tries by their correlation before costs. The probability is the deflated Sharpe ratio, corrected for the skewness (−0.3) and the kurtosis (10) of daily train returns.

9 Risks and limits

The universe is 22 survivors chosen in 2026, and the stress on the late-launched ETFs halves the Sharpe. The validation period, which lost money, is short and its range runs from −1.00 to 0.77, so it cannot be read as proof of decay or of noise. The daily returns have fat tails (kurtosis 9.96 on the train), and the result rests on a few days, many of them in the autumn of 2008. The author ran a version of this idea on the same ETFs in earlier work, so the train and the validation are not fresh to the human; only the test is. The Test note gives a full-sample t of 2.35 on monthly returns, while the daily figures in this report give 2.16. Both are above 2, but the choice of frequency moves it.

10 Decision

Passed, with a confirmed test: the test mean is 1.34 times the earlier mean, and the full-sample net t of 2.16 is above 2, so money is possible under the desk's rule, at half weight and only if the user adds it to the book. The edge is thin, the test t is 1.42, and the short leg loses, so this is a long-biased trend book. To revisit after twelve months of paper trading: if the net Sharpe on the new months stays under zero, or if the long leg stops paying, the strategy goes back to review.

A Monthly returns

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
2006	−0.0	−0.3	2.9	2.5	−2.9	0.2	−1.1	−1.7	−0.7	1.6	2.1	0.7	**3.3**
2007	0.9	−2.8	0.4	1.8	−0.5	−1.2	−1.6	−0.1	3.1	3.6	−0.1	1.1	**4.7**
2008	0.4	4.5	−0.7	−2.3	−0.8	4.7	−1.3	−1.3	3.0	3.5	3.1	−1.4	**11.4**
2009	1.3	1.4	−1.5	−2.0	−2.9	0.0	−1.0	−0.2	−0.1	−0.2	2.1	−0.8	**−4.0**
2010	−1.6	0.5	1.1	0.6	−3.5	0.9	0.8	3.2	1.5	2.9	−1.1	2.2	**7.3**
2011	0.1	2.2	0.4	3.6	−1.1	−1.2	1.8	−0.9	−5.5	−0.9	−0.7	0.6	**−1.6**
2012	0.4	−2.5	−0.4	1.1	4.7	−1.4	1.5	−1.4	−1.2	−0.3	0.7	1.9	**2.9**
2013	1.8	0.4	2.1	4.8	−2.7	−0.3	2.1	−1.6	3.0	1.4	2.0	2.5	**15.4**
2014	−3.0	−0.9	0.2	0.3	−0.7	0.7	−2.1	2.0	−1.4	4.3	3.3	2.2	**5.1**
2015	3.5	−0.2	2.1	−1.6	0.6	−2.2	4.6	−2.5	1.0	−1.9	4.0	0.8	**8.4**
2016	1.2	−2.3	−5.1	−1.2	−1.8	1.9	2.4	−1.5	0.5	−3.2	−2.5	−0.1	**−11.6**
2017	1.7	1.5	−0.1	0.3	0.7	−0.1	1.6	−0.4	1.6	1.5	1.5	1.7	**11.3**
2018	3.8	−3.5	−0.5	−0.7	−1.3	−0.9	0.9	0.8	1.6	−1.5	−1.7	−1.5	**−4.5**
2019	−5.6	0.1	1.9	−0.1	2.4	−1.4	1.9	5.5	−2.5	−1.4	0.3	0.7	**1.7**
2020	0.3	−1.8	3.1	0.3	−2.2	−0.8	−0.2	0.5	−1.4	0.2	−0.4	1.9	**−0.5**
2021	−0.0	−0.7	−0.2	1.8	1.5	−0.1	−0.4	0.8	−0.4	2.6	−2.6	3.5	**5.8**
2022	−1.0	−0.2	5.0	4.4	−1.2	4.3	−3.4	4.0	5.3	−0.1	−5.7	0.5	**12.0**
2023	−3.5	1.9	−1.9	−0.3	2.0	−1.3	0.8	−1.2	−1.6	−0.1	0.6	−1.0	**−5.6**
2024	0.4	2.1	3.5	−0.6	0.1	0.4	−1.3	0.1	1.4	−3.0	1.5	−2.7	**1.9**
2025	0.4	−1.7	−1.6	−0.1	0.2	1.7	−1.6	1.6	1.9	0.6	0.8	1.5	**3.6**
2026	2.1	1.3	−3.0	3.8	0.6	−0.4	−0.2	2.1	1.0				**7.5**

Table 10: Monthly net return, % of capital.

Runs: 20261109-091945-train-base, 20261109-103847-validation-base, 20261109-103855-train+validation-base, 20261109-112907-test-base. Strategy file: etf_tsmom_12m.py. Frozen version: base.